CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 33 (2009) 162–169

Contents lists available at ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic assessment of the H–La–Ni system

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ARTICLE INFO

Article history:
Received 29 May 2008
Received in revised form
4 August 2008
Accepted 3 September 2008
Available online 21 September 2008

Keywords:
H–La–Ni
Hydrogen storage
Phase diagram
Thermodynamics
Metallic hydrides

ABSTRACT

The H–La–Ni system has been assessed according to the CALPHAD method. To this purpose, an assessment of the La–Ni binary system has been taken from the literature, while the H–Ni system has been partially reassessed with respect to the liquid phase, in order to be consistent with generally used lattice stability of liquid hydrogen. A first assessment of binary H–La system has been carried out in this work. The binary LaH_x phase has been described with the sublattice model, in order to take into account a composition variations for $2 \leq x \leq 3$. Calculated p–c–T curves and Van't Hoff plots agree well with experimental data. In the ternary system, hydrogen absorption data have been considered for the $LaNi_5$ intermetallic compounds, leading to the $LaNi_5H_7$ hydride. Because the compound and the hydride have the same structure, the equilibrium between the two phases has been described as miscibility gap. Also in this case, calculated p–c–T curves and Van't Hoff plots agree rather well with experimental data. The equilibrium phases have been calculated in the ternary system. The present thermodynamic description can be a useful tool for predicting the hydrogen absorption and desorption behaviour in commercial vessels based on the AB_5 compounds.

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1. Introduction

Compounds of hydrogen and metallic elements have attracted attention in recent years because of possible applications for hydrogen storage. To this purpose, lanthanum–nickel compounds were already studied in the sixties and in the seventies. Among them, the $LaNi_5$ intermetallic compound has suitable features for commercial applications, apart for the high specific weight, and is currently used in commercial vessels for hydrogen storage.

Thermodynamic properties of the $LaNi_5$ compound and the related $LaNi_5H_7$ hydride are of great importance since they determine the pressure–temperature relationship for hydrogen absorption/desorption. A full understanding of the ternary H–La–Ni phase diagram is thus useful to assess the current experimental results obtained for lanthanum–nickel hydride and in order to better describe the reactions occurring in hydrogen absorption/desorption and related processes. However, to date no CALPHAD assessment of the binary H–La and of the ternary H–La–Ni systems have been reported in the literature.

The present paper will focus on the thermodynamic modelling of the most important systems for H–La–Ni hydrogen storage materials: La–Ni, H–La, H–Ni and H–La–Ni using the CALPHAD (CALculation of PHAse Diagrams) approach. La–Ni was already assessed by several authors [1–3]. The H–La and H–La–Ni systems are assessed in the present work. Furthermore, a partial reassessment of the H–Ni system has also been carried out in the present study. The model parameters were evaluated by using all available experimental and theoretical thermodynamic and phase diagram data. Section 2 will review literature information on these systems. In Section 3 thermodynamic models for each phase will be presented. In Section 4 results of thermodynamic assessment will be presented and discussed, followed by conclusions in Section 5.

2. Review of literature information

2.1. La–Ni system

The binary La–Ni system has been investigated by several authors and a number of CALPHAD assessments of the system are available in the literature [1–3]. The phase diagram is rather complex with 9 intermetallic compounds (La_3Ni, La_7Ni_3, LaNi, La_2Ni_3, La_7Ni_{16}, $LaNi_3$, αLa_2Ni_7, βLa_2Ni_7, $LaNi_5$), 3 solution phases (dhcp La, fcc La and Ni, bcc La) and the liquid phase. The first assessment reported by Du et al. [1] has considered all intermetallics as stoichiometric compounds, while fcc, bcc and hcp phases were considered as regular solutions with a high interaction parameter to account for practically nil solubility. Redlich–Kister polynomials were used for the liquid phase. The calculated phase diagram according to this assessment is shown in Fig. 1. In the assessment reported by Dischinger and Schaller [3] a higher number of parameters have been used with respect to the assessment by Du et al., while the agreement between experimental data and calculations was not significantly improved. Moreover, it was impossible to reproduce the published diagram using those parameters. Finally, the most recent assessment

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doi:10.1016/j.calphad.2008.09.003

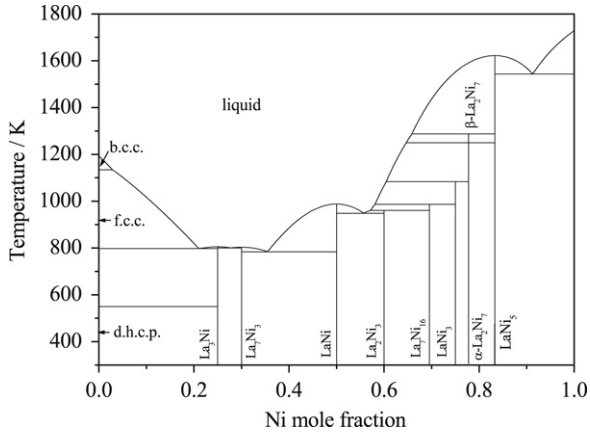

Fig. 1. Calculated La–Ni phase diagram according to the assessment by [1].

reported by Liu and Jin [2] basically reproduces the results by Du et al., with a slight improvement of the agreement between experimental and calculated enthalpies of formation of intermetallic compounds. A homogeneity range has been admitted for the $LaNi_5$ phase and a 3 sublattice model has been applied for its description implying significantly higher number of parameters with respect to the assessment by Du et al. [1].

2.2. H–Ni *system*

The H–Ni phase diagram is rather simple, since the only nickel hydride reported in the literature forms in a hydrogen atmosphere of more than 5000 bar [4], which is beyond the scope of the present work. The solubility of hydrogen in solid fcc and liquid Ni is limited at 1 bar as shown by the various experimental data reported in the literature. Moreover, data exist of hydrogen solubility at constant temperatures and different pressures, melting point of nickel in hydrogen atmosphere at different pressures, heats of solution of hydrogen in nickel. All experimental data are already summarized by Zeng et al. [5], who reported the only CALPHAD assessment available in the literature. Since in the present study we have adopted a different lattice stability for liquid hydrogen with respect to Zeng's assessment, a partial reassessment of the binary H–Ni system was necessary.

2.3. H–La *system*

The H–La phase diagram also shows a limited number of phases. The solubility of hydrogen in both fcc and bcc solid solutions is large, as well as in the liquid phase. One hydride is present in the system at ordinary pressures, LaH_x ($2 < x < 3$), which presents a wide range of homogeneity.

All experimental and theoretical information on the system has been collected by [6]. Phase diagram boundaries have been investigated for the fcc, bcc, LaH_x and liquid phase using thermal analysis, isothermal techniques and pressure–composition–temperature (p–c–T) measurements by a number of authors [7–13]. p–c–T measurements have been carried out at various temperatures, ranging from 473 to 1384 K. The enthalpy of formation of the LaH_x hydride has been determined calorimetrically by Dantzer and Kleppa [10]. Enthalpies and entropies for the hydrogenation reaction of fcc La have also been determined based on the use of Van't Hoff plots [7, 9,14], but these data correspond to average values in a certain temperature range. The entropy of LaH_x has been obtained on the base of specific heat measurements [15,16]. Thermochemical data on LaH_x are reported in Table 1. Experimental data are lacking in the H-rich side of the diagram (between LaH_x and hydrogen) and at high temperatures (above ∼1400 K). In the phase diagram two conflicting phase boundaries have been suggested by Peterson and Straatman [8] and Bischof et al. [12,14] at high temperature.

2.4. H–La–Ni *system*

Only one ternary phase has been reported in the system, i.e. the hydride $LaNi_5H_7$ obtained by hydrogen absorption in $LaNi_5$. Thus, most thermodynamic information on the ternary system regards the section $LaNi_5$–$LaNi_5H_7$, where several p–c–T measurements have been reported [17–28]. Thermodynamic data on the $LaNi_5H_7$ hydride have also been reported by a number of authors [17–21, 29–33] both by p–c–T measurements and calorimetry. They are summarized in Table 1.

Table 1
Calculated and experimental thermodynamic properties of hydrides in the H–La–Ni system

Compound	property	Reference phases H	La	Ni	T (K)	Calculated (J/mol of atoms)	Experimental (J/mol of atoms)	Notes	Reference
LaH_x	Enthalpy of formation	Gas	dhcp	–	917	−63 600	−63 700	Obtained by calorimetry	[10]
	Entropy of formation LaH_2	–	–	–	298.15	44.1	45	Obtained by Cp measurements	[15]
	Entropy of formation LaH_3	–	–	–	298.15	49.3	49.3	Obtained by Cp measurements	[16]
$LaNi_5H_x$	Enthalpy of formation	Gas	dhcp	fcc	298		−19 200[a]	Direct calorimetry involving the reaction of $LaNi_5$ with H_2	[29]
	Enthalpy of formation	Gas	dhcp	fcc	313–413		−18 800[a]	p–c–T	[20]
	Enthalpy of formation	Gas	dhcp	fcc	294–354		−19 000[a]	p–c–T	[17]
	Enthalpy of formation	Gas	dhcp	fcc	174–273		−20 100[a]	Calculated in Ref. [31] from absorption isotherms in Ref. [52]	[31]
	Enthalpy of formation	Gas	dhcp	fcc	296–345	−19 900	−18 800[a]	p–c–T	[18]
	Enthalpy of formation	Gas	dhcp	fcc	263–330		−19 600[a]	p–c–T	[19]
	Enthalpy of formation	Gas	dhcp	fcc	298.15		−19 400[a]	Obtained by heat capacities measured using adiabatic calorimetry	[31]
	Enthalpy of formation	Gas	dhcp	fcc	273.15		−19 600[a]		
	Enthalpy of formation	Gas	dhcp	fcc	195		−19 100[a]		
	Enthalpy of formation	Gas	dhcp	fcc	296		−19 300[a]	Obtained by calorimetry	[33]
	Enthalpy of formation	Gas	dhcp	fcc	285		−20 000[a]	Obtained by calorimetry	[32]
	Enthalpy of formation	Gas	dhcp	fcc	298		−19 700[a]	Obtained by calorimetry	[30]
	Enthalpy of formation	Gas	dhcp	fcc	298		−21 000[a]	Ab initio calculations, VASP	[36]

[a] Original data are for the hydrogenation reaction from the binary $LaNi_5$ compound. They have been converted into present values using thermodynamic assessed parameters for the $LaNi_5$ compound by [1].

Two ordered structures have been reported for the ternary hydride $LaNi_5H_7$ based on neutron diffraction data [34]. They have the $P6_3mc$ and P31c symmetries. Both unit cells correspond to doubling the unit cell of $LaNi_5$ along the *c*-axis and contain 14 H atoms at three different hydrogen sites. However, it is not yet clear which structure is the most stable and ab initio calculations recently carried out [35,36] have not solved the problem. Moreover, for the ternary hydride the same crystal symmetry of $LaNi_5$ with fractional occupancies of the hydrogen atoms has also been reported [37].

It must be pointed out that a different ternary hydride has been observed at high temperatures (above 340–360 K depending on absorption or desorption) in *p*–*c*–*T* measurements [25,38,39]. However, its structure is unknown to the authors knowledge and experimental data are limited. Thus, this phase was not included in the present assessment.

3. Modelling

In the framework of the CALPHAD methodology, the description of the Gibbs free energy of a phase φ is based on the following expression:

$$G^{\varphi} = {}^{\mathrm{ref}}G^{\varphi} + {}^{\mathrm{id}}G^{\varphi} + {}^{\mathrm{ex}}G^{\varphi} \tag{1}$$

where the first term (${}^{\mathrm{ref}}G^{\varphi}$) is the contribution of the pure components, the second one (${}^{\mathrm{id}}G^{\varphi}$) is the ideal mixing contribution and the third one is the contribution due to non-ideal interactions between the components, also known as the excess Gibbs energy of mixing. Each of these terms can depend on temperature, pressure and composition. Empirical parameters in them depend on the model adopted.

In the following, the pure solid elements in their stable phases at 298.15 K were chosen as reference state for the system (SER, Standard Element Reference) as recommended by SGTE (Scientific Group Thermodata Europe), except where otherwise specified. The thermodynamic functions of pure elements (lattice stabilities) have been taken from the database compiled by A. Dinsdale [40], except for the pure liquid hydrogen which is not included in the compilation. Two different lattice stabilities have been suggested in the literature for pure liquid hydrogen [5,41] based on extrapolation of binary and higher order data. In order to maintain consistency with most of published papers in the literature [41–44], we decided to use lattice stability according to these latter papers.

For condensed phases, the variation of the Gibbs free energy with pressure is negligible in the range of interest (<10 MPa) [45]. Thus, pressure dependence of the Gibbs energy was only taken into account for the gas phase.

3.1. The gas phase

The gas phase containing H, H_2, La and Ni species was considered as an ideal mixture. Moreover, an ideal gas behaviour has been assumed in the temperature and pressure range of interest. The molar Gibbs energy of the gas phase is then given by:

$$G^{\mathrm{gas}}_{\mathrm{H,H_2,La,Ni}}(T) = \sum_i x_i \left[{}^0G_i^{\mathrm{gas}} + RT\ln(P/P_0)\right] + RT\sum_i x_i \ln x_i \tag{2}$$

where ${}^0G_i^{\mathrm{gas}}$ is the standard Gibbs energy of the component *i* in the gas state, P_0 is the standard pressure of 101325 Pa, x_i is the mole fraction of species *i* and *R* is the gas constant. Diatomic H_2 is the dominant species, while the others have practically zero amounts.

3.2. The liquid phase

The binary liquid phase has been described by means of Redlich–Kister polynomials [46], while ternary interactions have been considered according to the Muggianu scheme [47]. Thus, the Gibbs free energy of the liquid phase is described as

$$\begin{aligned}
{}^{\mathrm{ref}}G^{\mathrm{liq}} &= x_{\mathrm{H}}G^{\mathrm{liq}}_{\mathrm{H}} + x_{\mathrm{La}}G^{\mathrm{liq}}_{\mathrm{La}} + x_{\mathrm{Ni}}G^{\mathrm{liq}}_{\mathrm{Ni}} \\
{}^{\mathrm{id}}G^{\mathrm{liq}} &= RT[x_{\mathrm{H}}\ln(x_{\mathrm{H}}) + x_{\mathrm{La}}\ln(x_{\mathrm{La}}) + x_{\mathrm{Ni}}\ln(x_{\mathrm{Ni}})] \\
{}^{\mathrm{ex}}G^{\mathrm{liq}} &= x_{\mathrm{H}}x_{\mathrm{La}}\sum_{\nu}{}^{\nu}L^{\mathrm{liq}}_{\mathrm{H,La}}(x_{\mathrm{H}}-x_{\mathrm{La}})^{\nu} \\
&\quad + x_{\mathrm{H}}x_{\mathrm{Ni}}\sum_{\nu}{}^{\nu}L^{\mathrm{liq}}_{\mathrm{H,Ni}}(x_{\mathrm{H}}-x_{\mathrm{Ni}})^{\nu} \\
&\quad + x_{\mathrm{La}}x_{\mathrm{Ni}}\sum_{\nu}{}^{\nu}L^{\mathrm{liq}}_{\mathrm{La,Ni}}(x_{\mathrm{La}}-x_{\mathrm{Ni}})^{\nu} + x_{\mathrm{H}}x_{\mathrm{La}}x_{\mathrm{Ni}}L^{\mathrm{liq}}_{\mathrm{H,La,Ni}}
\end{aligned} \tag{3}$$

where ν is the order of the term considered and ${}^{\nu}L^{\mathrm{liq}}_{\mathrm{H,La}}$, ${}^{\nu}L^{\mathrm{liq}}_{\mathrm{H,Ni}}$ and ${}^{\nu}L^{\mathrm{liq}}_{\mathrm{La,Ni}}$ are interaction parameters which have been taken as temperature dependent (see Appendix for details). $L^{\mathrm{liq}}_{\mathrm{H,La,Ni}}$ is a ternary interaction parameter which has been neglected in the present study due to insufficient experimental data.

3.3. Solution phases

The dhcp La phase shows practically no solubility both with Ni and H. Thus, no interaction parameters were considered. Similarly, no solubility of La in fcc Ni was reported and only one positive regular solution parameter was set in the La–Ni system to avoid solubility.

On the contrary, hydrogen shows a wide solubility range in both fcc and bcc La, while nickel is insoluble in both previous phases. However, no experimental data are available to identify interstitial locations of hydrogen in these phases. Thus, the Gibbs energy of solid solution phases has been described using the sublattice model, where hydrogen is assumed to fill interstitial sites without distinguishing between tetrahedral or octahedral sites. For the fcc phase, 2 sublattices have been used $(\mathrm{La,Ni})_1(\mathrm{H,Va})_1$ and the Gibbs free energy is described as follows

$$\begin{aligned}
{}^{\mathrm{ref}}G^{\mathrm{fcc}} &= {}^2y_{\mathrm{H}}({}^1y_{\mathrm{La}}{}^0G^{\mathrm{fcc}}_{\mathrm{La:H}} + {}^1y_{\mathrm{Ni}}{}^0G^{\mathrm{fcc}}_{\mathrm{Ni:H}}) \\
&\quad + {}^2y_{\mathrm{Va}}({}^1y_{\mathrm{La}}{}^0G^{\mathrm{fcc}}_{\mathrm{La:Va}} + {}^1y_{\mathrm{Ni}}{}^0G^{\mathrm{fcc}}_{\mathrm{Ni:Va}}) \\
{}^{\mathrm{id}}G^{\mathrm{fcc}} &= RT[{}^1y_{\mathrm{La}}\ln({}^1y_{\mathrm{La}}) + {}^1y_{\mathrm{Ni}}\ln({}^1y_{\mathrm{Ni}})] \\
&\quad + RT[{}^2y_{\mathrm{H}}\ln({}^2y_{\mathrm{H}}) + {}^2y_{\mathrm{Va}}\ln({}^2y_{\mathrm{Va}})] \\
{}^{\mathrm{ex}}G^{\mathrm{fcc}} &= {}^2y_{\mathrm{H}}{}^2y_{\mathrm{Va}}{}^0L^{\mathrm{fcc}}_{\mathrm{La:H,Va}} + {}^2y_{\mathrm{H}}{}^2y_{\mathrm{Va}}{}^1L^{\mathrm{fcc}}_{\mathrm{La:H,Va}}
\end{aligned} \tag{4}$$

where ${}^0G^{\mathrm{fcc}}_{\mathrm{La:H}}$, ${}^0G^{\mathrm{fcc}}_{\mathrm{Ni:H}}$, ${}^0G^{\mathrm{fcc}}_{\mathrm{La:Va}}$, ${}^0G^{\mathrm{fcc}}_{\mathrm{Ni:Va}}$ are the Gibbs free energy of the "end-members", ${}^jy_i$ is the site fraction of component *i* in sublattice *j* and ${}^0L^{\mathrm{fcc}}_{\mathrm{La:H,Va}}$, ${}^1L^{\mathrm{fcc}}_{\mathrm{La:H,Va}}$ are binary interaction parameters which are temperature dependent (see Appendix for details).

Analogously, a two sublattice model has been adopted for the bcc phase, but with a different ratio between sublattices $(\mathrm{La,Ni})_1(\mathrm{H,Va})_3$.

3.4. The binary hydride LaH_x

The binary LaH_x hydride has an fcc structure, with space group $Fm\bar{3}m$. Usually, La atoms occupy the 4a sites, while hydrogen atoms occupy the tetrahedral 8c and octahedral 4b sites. Experimental determinations using several different techniques [6] show that hydrogen atoms occupy tetrahedral sites first, for concentration up to $H/M = 1.95$, and only above this concentration value octahedral sites begin to be occupied. Accordingly, a 3 sublattice model has been applied as $(\mathrm{La})_1(\mathrm{H,Va})_2(\mathrm{H,Va})_1$, the second sublattice corresponding to tetrahedral sites while the third sublattice corresponds to the octahedral sites. The subsequent expression for the Gibbs energy of LaH_x is given by

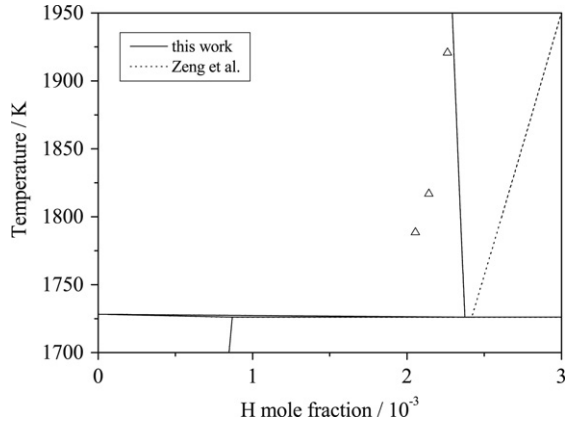

Fig. 2. Calculated H–Ni phase diagram according to Ref. [5] (dotted line) and to this work (continuous line). Experimental points from Ref. [5].

$$
\begin{aligned}
{}^{\mathrm{ref}}G^{\mathrm{LaH}_x} &= {}^2y_{\mathrm{H}}({}^3y_{\mathrm{H}}\,{}^0G^{\mathrm{LaH}_x}_{\mathrm{La:H:H}} + {}^3y_{\mathrm{Va}}\,{}^0G^{\mathrm{LaH}_x}_{\mathrm{La:H:Va}}) \\
&\quad + {}^2y_{\mathrm{Va}}({}^3y_{\mathrm{H}}\,{}^0G^{\mathrm{LaH}_x}_{\mathrm{La:Va:H}} + {}^3y_{\mathrm{Va}}\,{}^0G^{\mathrm{LaH}_x}_{\mathrm{La:Va:Va}}) \\
{}^{\mathrm{id}}G^{\mathrm{LaH}_x} &= 2RT[{}^2y_{\mathrm{H}}\ln({}^2y_{\mathrm{H}}) + {}^2y_{\mathrm{Va}}\ln({}^2y_{\mathrm{Va}})] \\
&\quad + RT[{}^3y_{\mathrm{H}}\ln({}^3y_{\mathrm{H}}) + {}^3y_{\mathrm{Va}}\ln({}^3y_{\mathrm{Va}})] \\
{}^{\mathrm{ex}}G^{\mathrm{LaH}_x} &= {}^2y_{\mathrm{H}}\,{}^2y_{\mathrm{Va}}\,{}^3y_{\mathrm{H}}\,{}^0L^{\mathrm{LaH}_x}_{\mathrm{La:H,Va:H}} + {}^2y_{\mathrm{H}}\,{}^2y_{\mathrm{Va}}\,{}^3y_{\mathrm{Va}}\,{}^0L^{\mathrm{LaH}_x}_{\mathrm{La:H,Va:Va}} \\
&\quad + {}^2y_{\mathrm{H}}\,{}^3y_{\mathrm{H}}\,{}^3y_{\mathrm{Va}}\,{}^0L^{\mathrm{LaH}_x}_{\mathrm{La:H:H,Va}} + {}^2y_{\mathrm{Va}}\,{}^3y_{\mathrm{H}}\,{}^3y_{\mathrm{Va}}\,{}^0L^{\mathrm{LaH}_x}_{\mathrm{La:Va:H,Va}}
\end{aligned}
\tag{5}
$$

where ${}^0G^{\mathrm{LaH}_x}_{\mathrm{La:H:H}}$, ${}^0G^{\mathrm{LaH}_x}_{\mathrm{La:Va:H}}$, ${}^0G^{\mathrm{LaH}_x}_{\mathrm{La:H:Va}}$, ${}^0G^{\mathrm{LaH}_x}_{\mathrm{La:Va:Va}}$ are the Gibbs free energy of the "end-members", ${}^jy_i$ is the site fraction of component i in sublattice j and ${}^0L^{\mathrm{LaH}_x}_{\mathrm{La:*:H,Va}}$, ${}^0L^{\mathrm{LaH}_x}_{\mathrm{La:H,Va:*}}$ are binary interaction parameters which can be temperature dependent (see Appendix for details).

3.5. *The ternary hydride* $LaNi_5H_7$

According to the structural information reported in Section 2.4 and the experimental data available on this phase, a 3 sublattice model (La)(Ni)$_5$(H, Va)$_7$ has been chosen for its Gibbs free energy. We have neglected the limited solubility range evidenced in the binary $LaNi_5$ by [2] in order to limit the number of parameters considered here. The corresponding equation is given by

$$
\begin{aligned}
{}^{\mathrm{ref}}G^{\mathrm{LaNi_5H}_x} &= {}^3y_{\mathrm{H}}\,{}^0G^{\mathrm{LaNi_5H}_x}_{\mathrm{La:Ni:H}} + {}^3y_{\mathrm{Va}}\,{}^0G^{\mathrm{LaNi_5H}_x}_{\mathrm{La:Ni:Va}} \\
{}^{\mathrm{id}}G^{\mathrm{LaNi_5H}_x} &= RT[{}^3y_{\mathrm{H}}\ln({}^3y_{\mathrm{H}}) + {}^3y_{\mathrm{Va}}\ln({}^3y_{\mathrm{Va}})] \\
{}^{\mathrm{ex}}G^{\mathrm{LaNi_5H}_x} &= {}^3y_{\mathrm{H}}\,{}^3y_{\mathrm{Va}}\,{}^0L^{\mathrm{LaNi_5H}_x}_{\mathrm{La:Ni:H,Va}}.
\end{aligned}
\tag{6}
$$

4. Results and discussion

According to the CALPHAD approach, the model parameters are determined by a computer-assisted statistical procedure in order to obtain the best fit of available experimental data based on numerical techniques [48]. In the assessment procedure, the differences between experimental and computed values are minimized by a least squares method. The optimised parameters are listed in the Appendix. The ThermoCalc software and its Parrot module [49] have been used for thermodynamic optimization.

4.1. *Partial reassessment of the* H–Ni *system*

As mentioned in the foregoing, in this study we have adopted a different lattice stability for pure liquid hydrogen with respect to that used in the assessment of the H–Ni system carried out by Zeng et al. [5]. Thus, we have reassessed the binary interaction parameters for the liquid phase according to our choice for the lattice stability of liquid hydrogen.

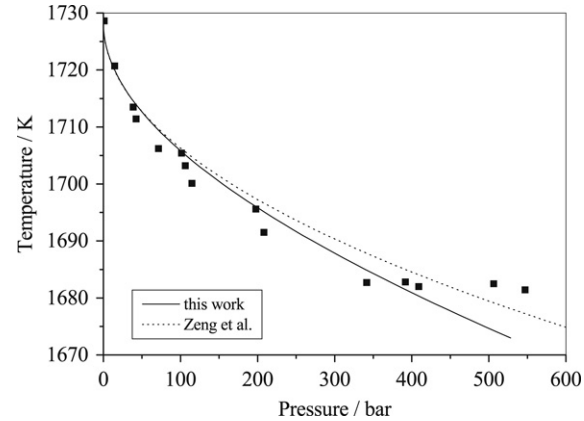

Fig. 3. Calculated melting point of H–Ni solution in hydrogen atmosphere at different pressures, according to Ref. [5] (dotted line) and to this work (continuous line). Experimental points from Ref. [5].

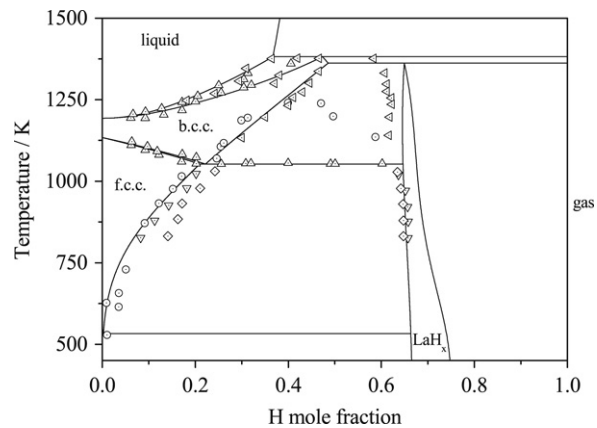

Fig. 4. Calculated H–La phase diagram (continuous line). Experimental points from Ref. [8] from isothermal equilibria (·), Ref. [8] from thermal analysis (9), Ref. [9] (Δ), Ref. [11] (N) and Ref. [12] (Ψ).

A summary of our results compared with previous results from Zeng et al. [5] and experimental findings is reported in Figs. 2 and 3. The agreement obtained between experimental data and CALPHAD calculations is satisfactory and comparable to the agreement obtained by Zeng et al. [5].

4.2. *Assessment of the binary* H–La *system*

The calculated phase diagram according to present assessment is reported in Fig. 4. A good agreement has been obtained between experimental data and calculated phase boundaries. The eutectoid point corresponding to the reaction bcc → fcc + LaH_x occurs at 1053 K, in good agreement with the experimental value equal to 1051 K [8]. The eutectoid point at lower temperature (fcc → dhcp + LaH_x) is also in satisfactory agreement with the limited experimental data. Phase boundaries are well described by present calculations. However, the high temperature region of the diagram involving equilibria between bcc and LaH_x phases deserves some comments. Two conflicting data sets have been reported in the literature by [8] and [12,14]. According to Peterson et al. [8], a miscibility gap occurs in the bcc phase at temperature in the range 1050–1233 K, the solubility range of H in the bcc phase above this critical point extends to high H concentrations and the LaH_x hydride should decompose into the bcc phase on heating. However, no experimental evidence of this transition has

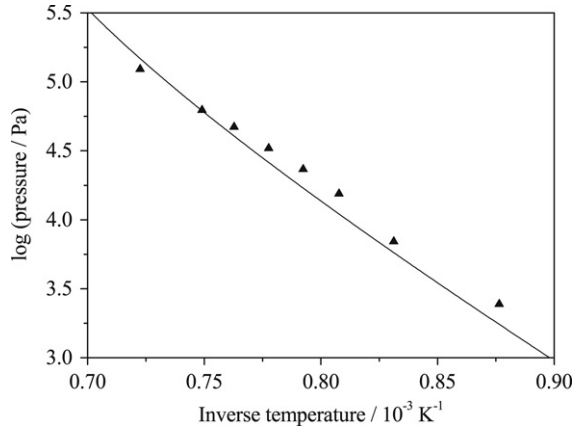

Fig. 5a. Calculated Van't Hoff plot for fcc phase in the H–La system (continuous line). Experimental points from Ref. [12].

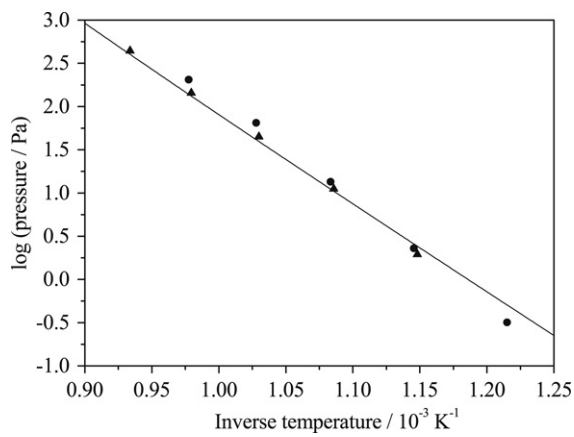

Fig. 5b. Calculated Van't Hoff plot for bcc phase in the La–H system (continuous line). Experimental points from Ref. [11] (,) and from Ref. [7] (7).

been found by the authors. Moreover, they recognized that it was difficult in their metallographic experiments to distinguish between the bcc and LaH_x phases at high hydrogen concentration. On the contrary, Bischoff et al. [12] found experimentally that the biphasic region extends to temperature up to 1384 K, the highest temperature measured. Thus, they conclude that the miscibility gap should exist at least up to this temperature or that equilibria between bcc and LaH_x phases should occur. They suggested that previous results by Peterson et al. could be wrong at these temperatures and also suggest the occurrence of a peritectic reaction bcc $\rightarrow$ liq + LaH_x. The present results confirm that no miscibility gap occurs in the bcc phase and equilibria between bcc and LaH_x phases are calculated up to 1363 K, where the decomposition of the binary hydride occurs into bcc and gas phase. At a slightly higher temperature (1382 K), bcc phase decomposes peritectically into liquid and gas phases. The difference between these two transition temperatures is however limited and further experimental investigations are desirable to confirm present results.

The calculated Van't Hoff plots for both the fcc and bcc phases are reported in Figs. 5a and 5b. Experimental points from *p–c–T* measurements are also reported for comparison. A good agreement has been obtained both for fcc and bcc phases in different temperature ranges. The calculated Van't Hoff plot of the bcc phase shows a more distinct bending, i.e. a significant temperature dependence of the enthalpy of hydrogenation reaction.

In Fig. 6, calculated *p–c–T* curves are compared with experimental determinations at several temperatures. Low temperatures

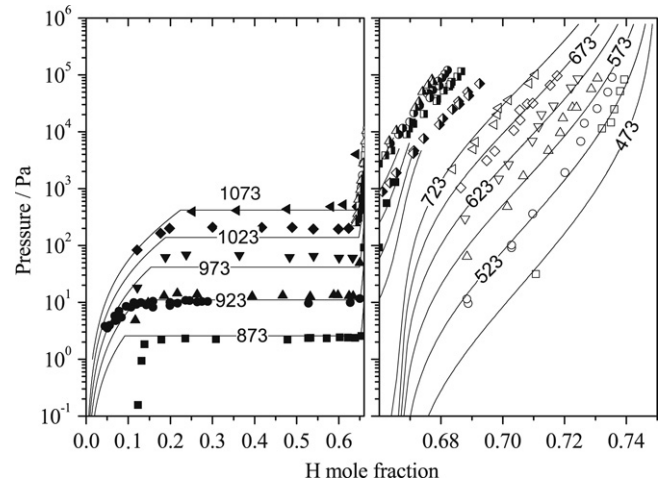

Fig. 6. Calculated *p–c–T* curves in the H–La system (continuous lines, temperature indicated in K). Experimental points from Ref. [11] at 873 K (!), from Ref. [10] at 917 K (,), from Ref. [11] at 923 K (7), from Ref. [11] at 973 K (B), from Ref. [11] at 1023 K (Λ), from Ref. [7] at 1071 K (Ω), from Ref. [7] at 929 K in desorption ()), from Ref. [7] at 979 K in desorption (4), from Ref. [7] at 1029 K in desorption (?), from Ref. [7] at 879 K in desorption (T), from Ref. [7] at 473 K in abs/des ($\forall$), from Ref. [7] at 523 K (–), from Ref. [7] at 573 K (8), from Ref. [7] at 623 K (X), from Ref. [7] at 673 K (M) and from Ref. [7] at 723 K (Ξ).

measurements (473–723 K) occur in the solubility range of LaH_x without involving the fcc La metal. Agreement between experimental and calculated results in this temperature range is satisfactory, with the highest deviation occurring at low temperature. Only above 823 K three phases equilibria among gas, fcc La and LaH_x are observed. The general agreement between calculations and experiments is good. Interestingly, some experimental data in the temperature range 873–1023 K are conflicting and experimental *p–c–T* curves seems to cross each other, which is thermodynamically impossible.

Finally, a very good agreement has been obtained between calculated and experimental thermochemical data, as evidenced in Table 1.

4.3. *Assessment of the* $LaNi_5$–$LaNi_5H_7$ *system*

A comparison between thermochemical experimental data on the ternary $LaNi_5H_7$ hydride and calculated results according to present assessment is reported in Table 1. Some of the experimental data have been obtained from *p–c–T* measurements and are thus given in a certain temperature range. Various sets of data show only limited differences among each other and calculated results fit well the experimental values. On the whole, the thermodynamic behaviour of the ternary hydride is well described by the present optimized parameters. The calculated value of the formation enthalpy shows a negligible temperature change since no specific heat terms has been considered in its thermodynamic modelling.

Fig. 7 reports the calculated Van't Hoff plot for the ternary $LaNi_5H_7$ hydride. Several data sets are also shown for comparison and the agreement obtained is good. It must be pointed out that *p–c–T* measurements are greatly affected by sample preparation/conditions/activation. For example, the first absorption/desorption cycle always gives a *p–c–T* curve significantly different from the following ones. During optimization and in Fig. 7 a selection of *p–c–T* data has been used, whose conditions are closer to equilibrium.

It must be pointed out that according to present thermodynamic description the ternary $LaNi_5H_7$ hydride is metastable with respect to the mixture of f.c.c. Ni, LaH_x and H_2 gas phases at room temperature and 1 bar pressure. However, a decomposition of the

Table A.1
Summary of optimized thermodynamic parameters for the H–La–Ni system

Gas phase, ideal mixture of H2, H, LA, NI, NI2. *Only* H_2 *description is given here.*			
G(GAS, H2; 0) = GH2GAST + $RT\ln(P)$			
Liquid phase, model Redlich–Kister–Muggianu (H, La, Ni)			
G(LIQUID,LA;0)	$+5332.653 + 18.23012 * T - 11.0188191 * T * \ln(T) - .020171603 * T^2 + 2.93775E - 06 * T^3 - 133\,541 * T^{(-1)}$	$298 < T < 1134$	From Ref. [40]
	$-3942.004 + 171.018431 * T - 34.3088 * T * \ln(T)$	$1134 < T < 4000$	
G(LIQUID,NI;0)	$+11\,235.527 + 108.457 * T - 22.096 * T * \ln(T) - .0048407 * T^2 - 3.82318E - 21 * T^7$	$298 < T < 1728$	From Ref. [40]
	$-9549.775 + 268.598 * T - 43.1 * T * \ln(T)$	$1728 < T < 3000$	
G(LIQUID,H;0)	$+8035 + 25 * T + 2 * T * \ln(T) + 0.5 *$ GH2GAST	$298 < T < 6000$	From Ref. [44]
G(LIQUID,H,LA;0)[a]	$+1.5304693E + 04 - 8.9375563E + 01 * T$	$298 < T < 6000$	
G(LIQUID,H,LA;1)[a]	$6.5990058E + 04 - 7.7403240E + 01 * T$	$298 < T < 6000$	From Ref. [5]
G(LIQUID,H,NI;0)	$-8.69093694E + 03 + 1.07429419E + 01 * T$	$298 < T < 6000$	
G(LIQUID,LA,NI;0)	$-116\,299 + 19.815 * T$	$298 < T < 3000$	From Ref. [1]
G(LIQUID,LA,NI;1)	$+63\,813 - 40.941 * T$	$298 < T < 3000$	From Ref. [1]
Bcc phase, model $(La, Ni)_1(H, Va)_3$			
G(BCC_A2,LA:VA;0)	$-3952.161 + 88.072353 * T - 21.7919 * T * \ln(T) - .004045175 * T^2 - 5.25865E - 07 * T^3$	$298 < T < 800$	
	$+321\,682.673 - 3565.08252 * T + 513.440708 * T * \ln(T) - .387295093 * T^2 + 4.9547989E - 05 * T^3 - 36581\,228 * T^{(-1)}$	$800 < T < 1134$	From Ref. [40]
	$-16\,377.894 + 218.492988 * T - 39.5388 * T * \ln(T)$	$1134 < T < 1193$	
	$-136\,609.91 + 1123.34397 * T - 163.413074 * T * \ln(T) + .053968535 * T^2 - 4.056395E - 06 * T^3 + 21167204 * T^{(-1)}$	$1193 < T < 2000$	
	$-8205.988 + 174.836315 * T - 34.3088 * T * \ln(T)$	$2000 < T < 4000$	
G(BCC_A2,LA:H;0)[a]	$-1.6628419E + 05 + 1.4021312E + 02 * T +$ GHSERLA $+ 1.5 *$ GH2GAST	$298 < T < 6000$	
G(BCC_A2,NI:VA;0)	$+3535.925 + 114.298 * T - 22.096 * T * \ln(T) - .0048407 * T^2$	$298 < T < 1728$	From Ref. [40]
	$-19\,125.571 + 275.579 * T - 43.1 * T * \ln(T) + 1.12754E + 31 * T^{(-9)}$	$1728 < T < 3000$	
TC(BCC_A2,NI:VA;0)	575	$298 < T < 3000$	From Ref. [40]
BMAGN(BCC_A2,NI:VA;0)	0.85	$298 < T < 3000$	From Ref. [40]
G(BCC_A2,LA,NI:VA;0)	$+100\,000$	$298 < T < 3000$	From Ref. [1]
Dhcp phase, model (La,Ni)			
G(DHCP,LA;0)	+GHSERLA	$298 < T < 4000$	From Ref. [40]
G(DHCP,NI;0)	+GHSERNI + 5000	$298 < T < 4000$	From Ref. [1]
G(DHCP,LA,NI;0)	$+100\,000$	$298 < T < 4000$	From Ref. [1]
Fcc phase, model $(La, Ni)_1(H, Va)_1$			
G(FCC_A1,LA:VA;0)	$-6109.797 + 89.878761 * T - 21.7919 * T * \ln(T) - .004045175 * T^2 - 5.25865E - 07 * T^3$	$298 < T < 1134$	From Ref. [40]
	$-124\,598.976 + 955.878375 * T - 139.346741 * T * \ln(T) + .042032405 * T^2 - 3.066199E - 06 * T^3 + 20\,994\,153 * T^{(-1)}$	$1134 < T < 2000$	
	$-12\,599.386 + 178.54399 * T - 34.3088 * T * \ln(T)$	$2000 < T < 4000$	
G(FCC_A1,LA:H;0)[a]	$-5.8190735E + 04 + 3.7002222E + 01 * T +$ GHSERLA $+ .5 *$ GH2GAST	$298 < T < 4000$	From Ref. [40]
G(FCC_A1,NI:VA;0)	+GHSERNI	$298 < T < 3000$	
TC(FCC_A1,NI:VA;0)	633	$298 < T < 3000$	From Ref. [40]
BMAGN(FCC_A1,NI:VA;0)	0.52	$298 < T < 3000$	From Ref. [40]
G(FCC_A1,LA,NI:VA;0)	$+100\,000$	$298 < T < 3000$	From Ref. [1]
G(FCC_A1,LA:H,VA;0)[a]	$3.9875816E + 03$	$298 < T < 3000$	From Ref. [5]
G(FCC_A1,NI:H;0)	$-3120.7324 + 180.5062 * T - 16.2550 * T * \ln(T) + 6.052879E - 4 * T^2 +$ GHSERNI $+ 0.5 *$ GH2GAST#	$298 < T < 3000$	
La_3Ni_1 *phase, model* $(La)_3(Ni)_1$			
G(LA3NI1,LA3NI1;0)	$-60\,909 - 0.702 * T + 3 *$ GHSERLA + GHSERNI	$298 < T < 6000$	From Ref. [1]
La_7Ni_3 *phase, model* $(La)_7(Ni)_3$			
G(LA7NI3,LA7NI3;0)	$-180\,100 + 1.054 * T + 7 *$ GHSERLA $+ 3 *$ GHSERNI	$298 < T < 6000$	From Ref. [1]
La_1Ni_1 *phase, model* $(La)_1(Ni)_1$			
G(LA1NI1,LA1NI1;0)	$-51\,390 - 0.055 * T +$ GHSERLA + GHSERNI	$298 < T < 6000$	From Ref. [1]
La_2Ni_3 *phase, model* $(La)_2(Ni)_3$			
G(LA2NI3,LA2NI3;0)	$-138\,509 + 14.109 * T + 2 *$ GHSERLA $+ 3 *$ GHSERNI	$298 < T < 6000$	From Ref. [1]
La_7Ni_{16} *phase, model* $(La)_7(Ni)_{16}$			
G(LA7NI16,LA7NI16;0)	$-639\,390 + 91.778 * T + 7 *$ GHSERLA $+ 16 *$ GHSERNI	$298 < T < 6000$	From Ref. [1]
La_1Ni_3 *phase, model* $(La)_1(Ni)_3$			
G(LA1NI3,LA1NI3;0)	$-108\,154 + 15.857 * T +$ GHSERLA $+ 3 *$ GHSERNI	$298 < T < 6000$	From Ref. [1]
α-La_2Ni_7 *phase, model* $(La)_2(Ni)_7$			
G(LA2NI7_alpha,LA2NI7;0)	$-239\,869 + 37.356 * T + 2 *$ GHSERLA $+ 7 *$ GHSERNI	$298 < T < 6000$	From Ref. [1]
β-La_2Ni_7 *phase, model* $(La)_2(Ni)_7$			
G(LA2NI7_beta,LA2NI7;0)	$-237\,259 + 35.267 * T + 2 *$ GHSERLA $+ 7 *$ GHSERNI	$298 < T < 6000$	From Ref. [1]
LaH_x *phase, model* $(La)_{0.25}(H, Va)_{0.5}(H, Va)_{0.25}$			
G(LAH3,LA:H:H;0)[a]	$-5.9306309E + 04 + 4.9300000E + 01 * T + .25 *$ GHSERLA $+ .375 *$ GH2GAST	$298 < T < 6000$	
G(LAH3,LA:VA:H;0)[a]	$8.8777368E + 03 + 9.9901014E - 01 * T + .25 *$ GHSERLA $+ .125 *$ GH2GAST	$298 < T < 6000$	
G(LAH3,LA:H:VA;0)[a]	$-4.8045126E + 04 + 3.3074723E + 01 * T + .25 *$ GHSERLA $+ .25 *$ GH2GAST	$298 < T < 6000$	
G(LAH3,LA:VA:VA;0)[a]	$8.8777368E + 03 + 9.9901014E - 01 * T + .25 *$ GHSERLA	$298 < T < 6000$	
G(LAH3,LA:H:H,VA;0)[a]	$-1.0446741E + 03$	$298 < T < 6000$	

Table A.1 (*continued*)

G(LAH3,LA:VA:H,VA;0)[a]	$-1.0446741E+03$	$298 < T < 6000$	
$LaNi_5H_7$ *phase, model* $(La)_{0.07692308}(Ni)_{0.38461538}(H, Va)_{0.53846154}$			
G(LANI5H7,LA:NI:VA;0)	$-11\,898 + 2.3555 * T + 0.07692308 * GHSERLA + 0.38461538 * GHSERNI$	$298 < T < 6000$	From Ref. [1]
G(LANI5H7,LA:NI:H;0)[a]	$-2.0575310E+04 + 3.2902621E+01 * T + 0.07692308 * GHSERLA + 0.38461538 * GHSERNI + 0.26923077 * GH2GAST$	$298 < T < 6000$	
G(LANI5H7,LA:NI:H,VA;0)[a]	$+2.1471913E+01 * T$	$298 < T < 6000$	
G(LANI5H7,LA:NI:H,VA;2)[a]	$+1.5353994E+03 - 2.4205966E+01 * T$	$298 < T < 6000$	
Functions and symbols			
GHSERLA	$-7968.403 + 120.284604 * T - 26.34 * T * \ln(T) - .001295165 * T^2$	$298 < T < 550$	
	$-3381.413 + 59.06113 * T - 17.1659411 * T * \ln(T) - .008371705 * T^2 + 6.8932E-07 * T^3 - 399\,448 * T^{(-1)}$	$550 < T < 2000$	From Ref. [40]
	$-15\,608.882 + 181.390071 * T - 34.3088 * T * \ln(T)$	$2000 < T < 4000$	
GHSERNI	$-5179.159 + 117.854 * T - 22.096 * T * \ln(T) - .0048407 * T^2$	$298 < T < 1728$	From Ref. [40]
	$-27\,840.655 + 279.135 * T - 43.1 * T * \ln(T) + 1.12754E+31 * T^{(-9)}$	$1728 < T < 3000$	
GH2GAST	$-9522.97393 + 78.5273873 * T - 31.35707 * T * \ln(T) + .0027589925 * T^2 - 7.46390667E-07 * T^3 + 56582.3 * T^{(-1)}$	$298 < T < 1000$	From Ref. [40]
	$+180.10884 - 15.6128262 * T - 17.84857 * T * \ln(T) - .00584168 * T^2 + 3.14618667E-07 * T^3 - 1280\,036 * T^{(-1)}$	$1000 < T < 2100$	
	$-18\,840.1661 + 92.3120249 * T - 32.05082 * T * \ln(T) - .0010728235 * T^2 + 1.14281783E-08 * T^3 + 3561\,002.5 * T^{(-1)}$	$2100 < T < 6000$	

[a] Parameters assessed in this work.

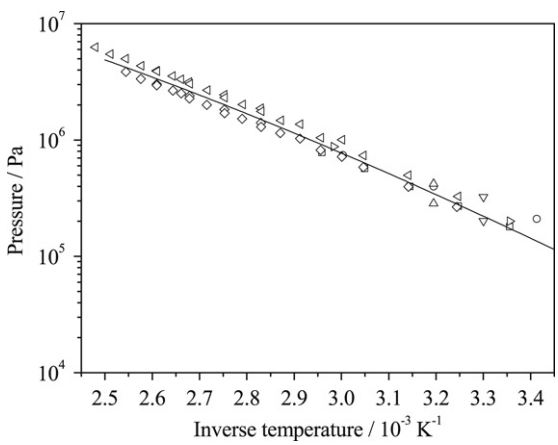

Fig. 7. Calculated Van't Hoff plot for $LaNi_5$ compound (continuous line). Experimental points from Ref. [28] (∀), from Ref. [50] (–), from Ref. [51] (8), from Ref. [22] (X), from Ref. [27] desorption (M), from Ref. [27] absorption (Ξ) and from Ref. [21] (χ).

ternary hydride has not been observed experimentally. In order to obtain a stable ternary hydride, the enthalpy of hydrogenation should be lower of about 4 kJ/mol of atoms. Thermochemical data of hydrogenation of $LaNi_5$ into the ternary hydride are nonetheless well established and different data sets are in reciprocal agreement. Calculations according to the present thermodynamic description are also in agreement with these experimental data. Since these data rely on the enthalpy of formation of the $LaNi_5$ phase, an underestimation of its value, calculated from Ref. [1], would also affect the stability of the ternary hydride. Experimental data on the enthalpy of formation of $LaNi_5$ are in fact scattered [1,3] and the lowest experimental values are 3–4 kJ/mol lower than the value assessed by Du et al. Thus a revision of the binary La–Ni system could significantly change the present results.

Another possibility is that the enthalpy of formation of the LaH_x hydride is overestimated in the present assessment. However, since the present assessment is in very good agreement with available experimental information, new experimental data would be necessary to further ascertain this possibility.

5. Conclusions

A thermodynamic description of the binary H–La and ternary H–La–Ni system have been obtained in the present work by modelling the Gibbs free energy of each phase. Experimental data have been collected from the literature, critically evaluated and used in the optimization procedure. The optimized binary interaction parameters for the liquid phase in the H–Ni system have been reoptimized in this work because of our different choice for the lattice stability of pure liquid hydrogen with respect to the previous assessment of the system. A good agreement between calculated and experimental phase diagram and thermochemical data in the binary H–La and ternary H–La–Ni has been obtained. According to present description, no miscibility gap occurs at high temperature in the bcc phase in the binary H–La phase diagram.

Acknowledgments

The EU-RTN "COSY" and Regione Piemonte "Innovative Materials for Hydrogen Storage" projects are acknowledged for supporting this work.

Appendix

See Table A.1.

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